Exam Date & Time: 22-Feb-2023 (02:30 PM - 05:30 PM)

# MANIPAL ACADEMY OF HIGHER EDUCATION

**FIRST SEMESTER B.TECH. EXAMINATIONS - FEBRUARY/MARCH 2023**
**SUBJECT: BIO 1071 / BIO_1071 - BIOLOGY FOR ENGINEERS (MAKEUP)**

**BIOLOGY FOR ENGINEERS [BIO 1071]**

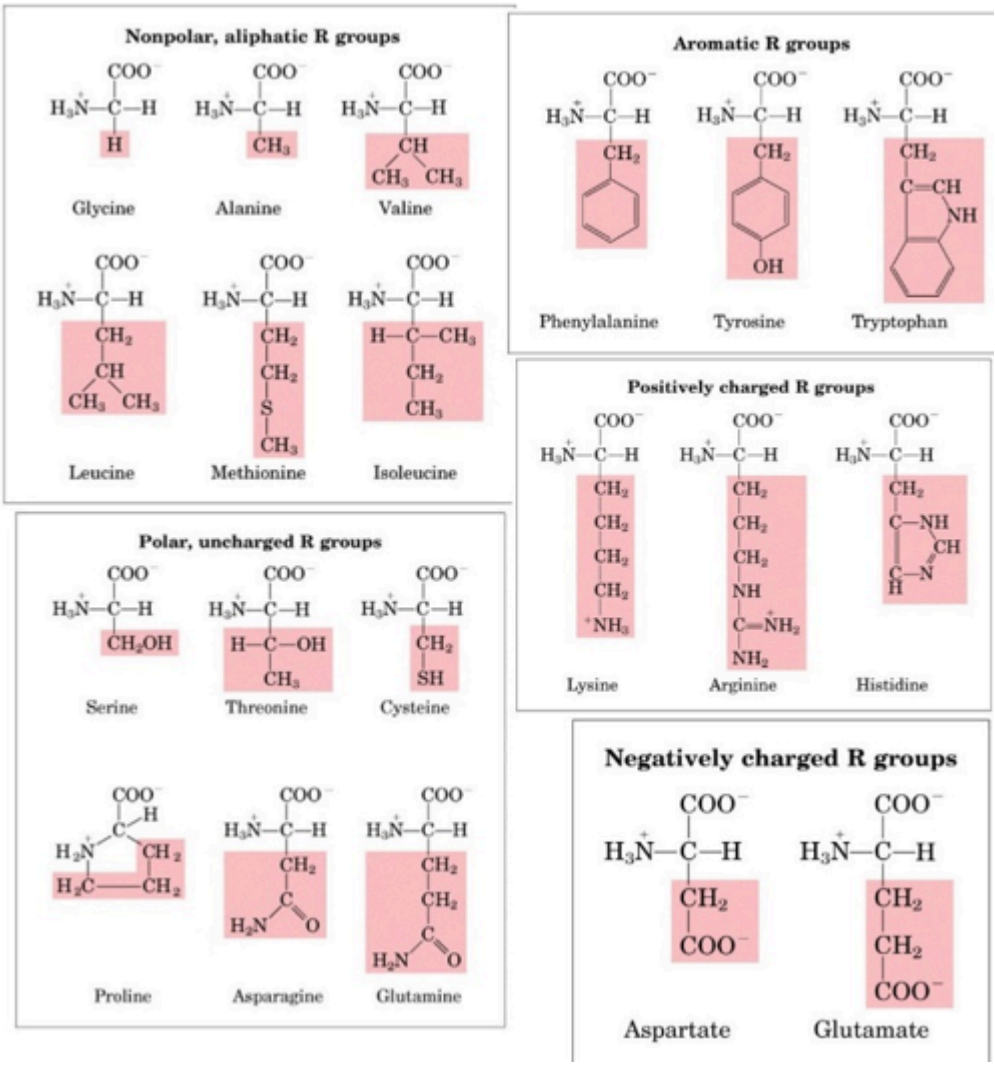
**Marks: 50** **Duration: 180 mins.**

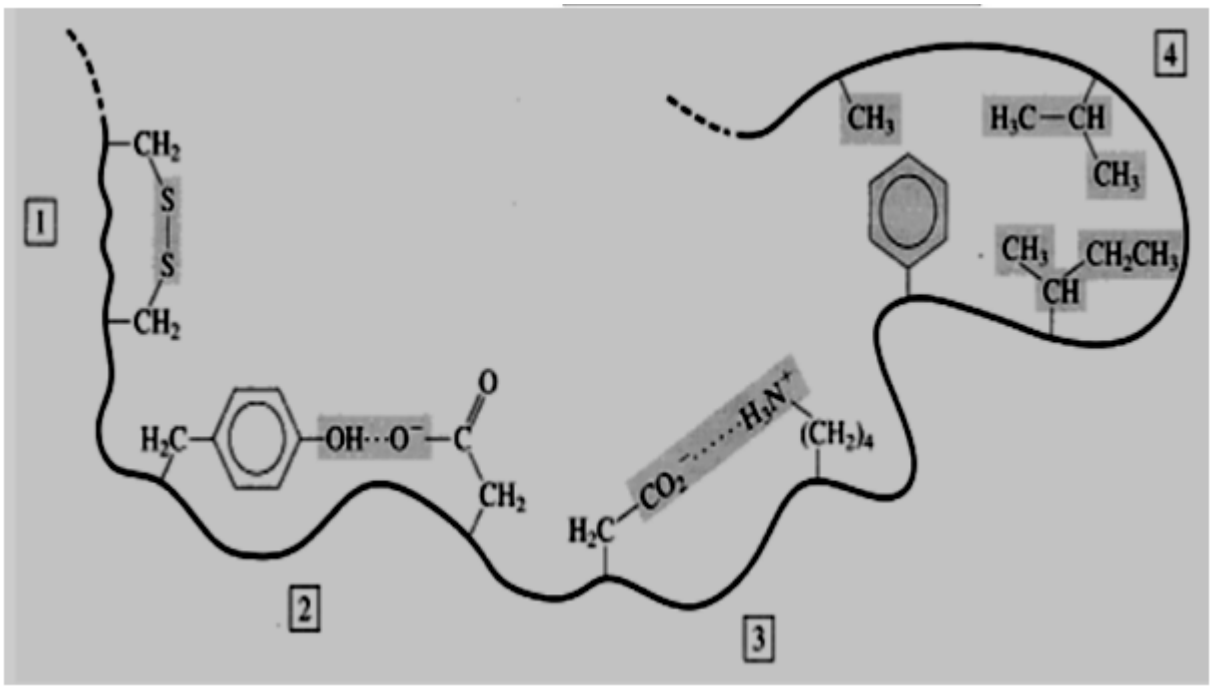
**A**

**Answer all the questions.**

1A) Illustrate the arrangement of phospholipids after placing them in water? Will they orient in the same way if they placed in oil? Give illustration of the arrangement. Justify (3)

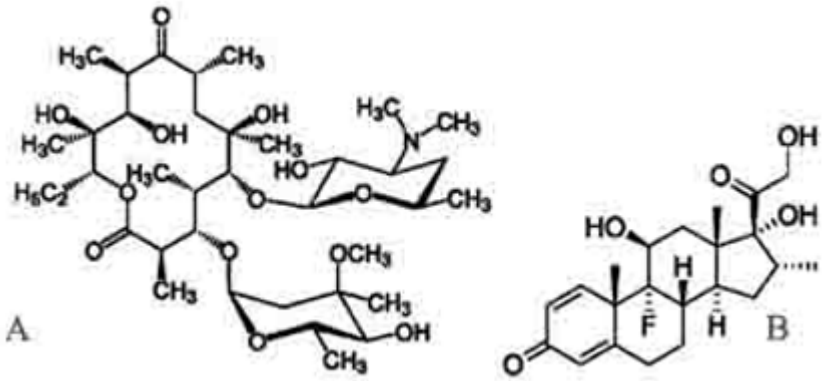
1B) Answer the following questions based on the given figure. (3)






i) Which structural level does the protein represent? (Choose from primary, secondary, tertiary, quaternary).
ii) Label the interactions mentioned in the figure 1, 2, 3 and 4.
iii) What will happen if hydrophobic amino acids in the part 4 of the protein were modified to serine?
(½+2+½ = 3 marks)

1C) Two different structures are given here. One is partially water-soluble antimicrobial drug erythromycin and the other one a corticosteroid (A) called dexamethasone (B). If both of them put in a common solvent at neutral pH, how do they interact with each other.

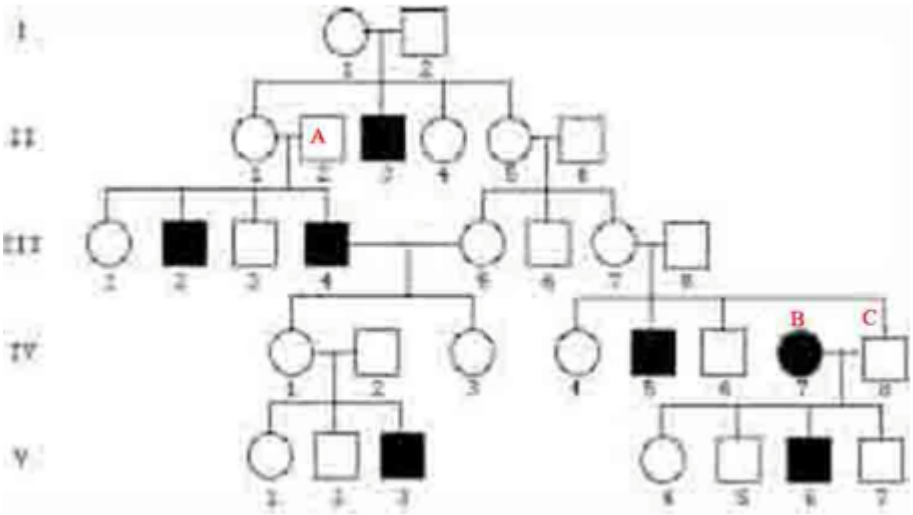



(4)

i) Can they make ionic bond with each other?
ii) Can they make vander Waal's force of interaction?
iii) Is it possible Hydrogen bonding between the two drugs?
iv) Both these compounds are very hydrophilic? Justify

2A) (3)

Analyse the following pedigree




i) Copy the following table and fill it with correct reasoning.

| Mode of inheritance | Yes/No with reason |
|---|---|
| Autosomal dominant | |
| X linked dominant | |
| X linked recessive | |
| Autosomal recessive | |

ii) Write the genotype of Individual A.
iii) Calculate the probability of the fifth child be affected for the parents B and C.
(2+½+½ = 3 marks)

2B) Ms Kumari conducted a typical Mendelian cross between two pea plants bearing green round (Dominant) and yellow wrinkled (Recessive) seeds. The F1 is crossed with green wrinkled and the outcome is photographed as below. YR (Yellow Round) and GR (Green Round seeds) are marked in the figure. Other phenotypes are self explanatory in the figure. (3)

i) Construct a Punnett square explaining the cross.
ii) What is the reason for obtaining yellow round and green wrinkled seeds?
iii) Calculate recombination frequency.
iv) Based on the results, predict the factors for seed colour and seed shape are located on

same chromosome or different chromosome?
(1+½+1+½ = 3 marks)

2C) i) Given below are four assertions. Prepare a correct reasoning explaining the assertion.

| Assertion | Reasoning |
|---|---|
| All crosses doesn't follow Mendelian pattern of inheritance | |
| Low recombination rate between two phenotypes | |
| Morgan's cross yielded a different outcome compared to Mendel's outcome | |

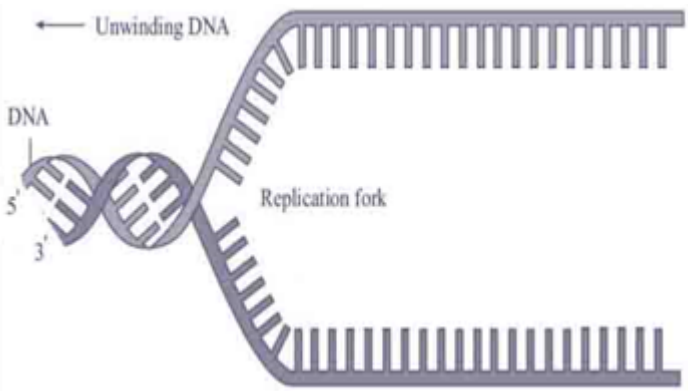
ii) In pea plants, inflated pod is governed by a completely dominant factor and constricted pod by a recessive factor. Another independently segregating dominant factor produces green pod while recessive factor produces yellow pod. Copy the table and give the genotypes of the parents in each of the crosses given below.

| Parental Phenotypes | Parental genotypes | Phenotypes of Offspring | | | |
|---|---|---|---|---|---|
| | | Inflated green | Inflated yellow | Constricted Green | Constricted yellow |
| inflated green x inflated yellow | _____ _____ | 58 | 62 | 18 | 22 |
| constricted green x constricted green | _____ _____ | 0 | 0 | 56 | 18 |
| inflated yellow x inflated yellow | _____ _____ | 0 | 64 | 0 | 20 |
| inflated green x inflated yellow | _____ _____ | 36 | 38 | 0 | 0 |
| inflated green x inflated green | _____ _____ | 178 | 62 | 58 | 22 |

(4)

(1½+2½ = 4 marks)

3A)



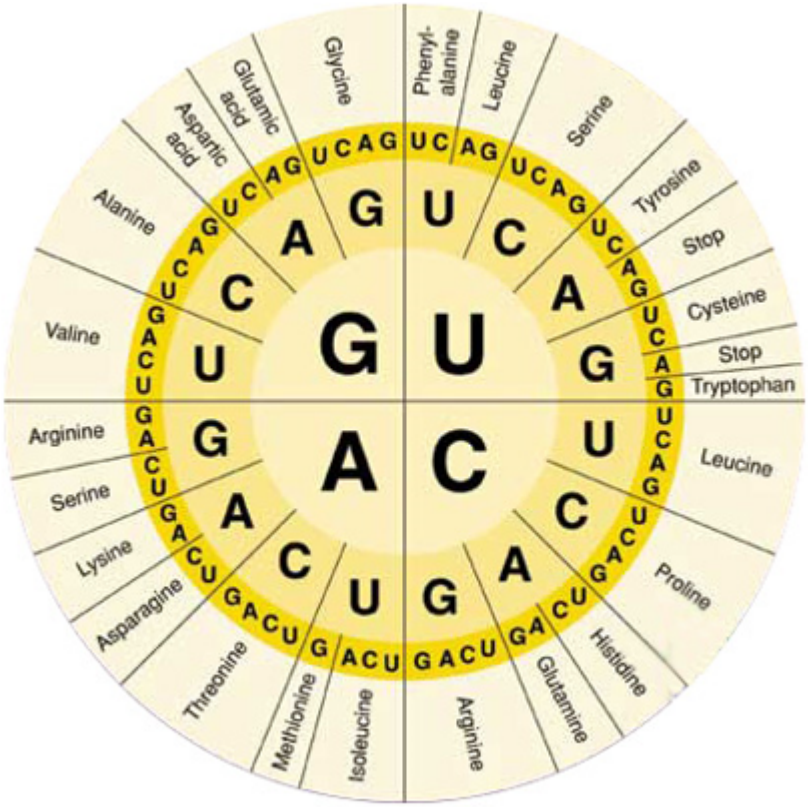

(3)

Fill the replication fork with leading and lagging strands, label the ends and label the location of primers.
What is the role of helicase and polymerase in this replication?

3B) Design an experiment to prove semi conservative mode of replication. (3)

3C) Complete the following table with the most suitable answer (4)

| Sl No | Template strand DNA Sequence with ends | mRNA sequence with ends | Amino acid |
|---|---|---|---|
| 1 | 5'ATG3' | | |
| 2 | | 5'UAA3' | |
| 3 | | | Methionine |
| 4 | | 5'UUU3' | |




4A) One of the classical examples of evolution occurs on the Galapagos Islands with Darwin finches. The islands have always been separating from the south American mainland and vary in size and elevation. The lowlands are covered with thorn scrub, while higher elevations are covered with moist and dense forests. All the organisms living on these islands are descendants of species that have migrated there, primarily from south America. In studying the finch populations, researchers have identified fourteen species, none of which are found on the mainland.

i) As opposed to the mainland, on the islands the initial flinches had the opportunity to evolve in new directions. Why? (3)

ii) The initial finch species on the islands may have all been seed eaters, which process minimized competition as the population explained?

iii) What evidence did scientists use to determine that all fourteen species of finches on the Galapagos have one common ancestor?

(1+1+1 = 3 marks)

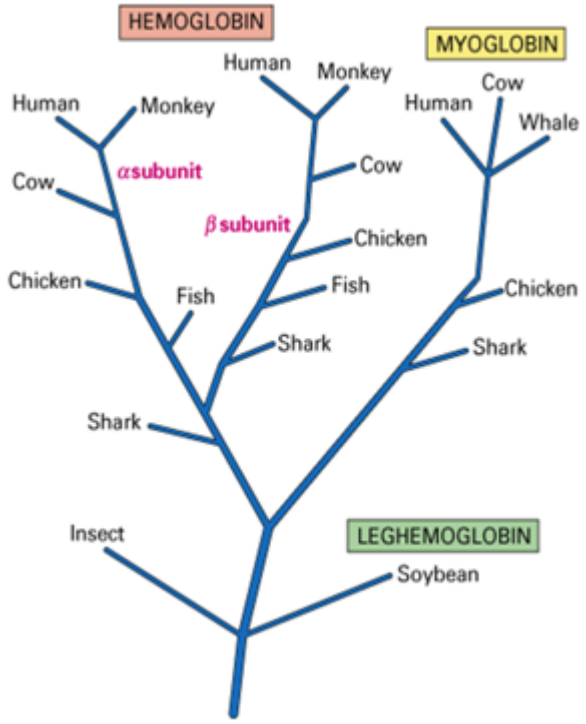
4B) Increased frequency of sickle cell anaemia in Africans was one of the relevant pieces of evidence of natural selection process. Sickle cell anaemia is an autosomal recessive trait.

i) Why the carrier of the sickle cell gene individuals lived to reproduced in African population? (3)

ii) What mutation is caused in sickle cell anaemia?

iii) Sickle cells are homozygous or heterozygous, why?

4C) The gene family encoding the globins is a good example of the gene families found in vertebrates. Answer the following questions based on the Phylogenetic tree showing the evolutionary history of globin proteins. (4)




i) If both are oxygen binding proteins, then what made the necessity of two oxygen binding molecules as a result of evolutionary process?
ii) The presence of pseudogenes is evidence of evolution in the gene, what is pseudogene?
iii) What is the function of Leghaemoglobin which is found in root nodules of legume plants like soyabean?
(2+1+1 = 4 marks)

5A) A DNA sequence undergoes three subsequent point mutation which results in subsequent changes in the transcription of mRNA as shown below:
Original mutation: AUG CAU CUC ACG GAU UAG
Point mutation 1: AUG CAU CUU ACG GUA UAG
Point mutation 2: AUG CAU GCU UAC GGU AUA
Point mutation 3: AUG CAU GCU UAA GGU AUA
Identify the type of mutation in the three sets. (3)

5B) Mention which type of interaction will be followed in the given statement?
i) A relationship where two organisms of species live in close physiological contact with each other.
ii) The cleaners eat insect's pests from the skin of animals.
iii) Sucker fish attached to shark for the transportation.
(1+1+1 = 3 marks) (3)

5C) Name bioinspired technologies for given examples:
i) Chameleon
ii) Gecko
iii) Australian thorny devil lizard
iv) Spider (4)

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